

movie film, possible first 225
 movie studio, first 228
 mp3 218, 219
 file name extension 330
 format software 329
 portable player 333
 MRI [magnetic resonance imaging]
 79, 278, 302
 functional 324
 mRNA 285
 MS-DOS 313
 Muhammad ibn Zakariya al-Razi
 [Rhazes] 43, 47, **48**, 368, 373
 Mulder, Gerardus 193
 mule, Crompton's 154
 Müller, Hermann 273
 Müller, Johannes 70
 Müller, Paul 271
 Mullis, Kary 318, 319
 multicylinder music box 218
 multiocular microscope 115
 mummification 23, 78
 Münchhausen syndrome 281
 muons 269
 Murchison, Roderick Impey 191,
 372
 Murdoch, Colin 91
 Murdoch, William 63, 161
 Murray, Joseph 283
 muscle contraction 117
 music box 218
 music player, iPod 338
 music therapy 48
 Musical Instrument Digital
 Interface 314, 315
 musical notation 51
 musical sound 104
 mutation
 and evolution 204
 de Vries 374
 Muybridge, Eadweard 217
 biography 372
Mycobacterium leprae 215
Mycoplasma mycoides 347
 myxomatosis 280, 281

N

NACA 252
 Nadler, Henry 296
 Nagaoka, Hantaro 239
 Nagasaki 275
 Nagle, Matthew 340
 Nakamura, Shuji **331**, 372
 nanotechnology 63
 nanotubes 261
 carbon 325
 Napier, John 98, 184
 biography 372
 Napier's bones 98, 184, 372
 NASA, established 288
 Nasmyth, James 196
 Nastulus 48
 Nathans, Daniel 301
 natron 23
 natural history 118
 Buffon 369
Natural History 36
 natural philosophy 96
 natural selection 178, 204, 206
Nature journal, founding of 211
 nautical log 133
Nautila 292–93
 nautilus 135
 navigation
 and Cortés 89
 tools for **132–33**
 world **71**
 NAVSTAR 308
 Nazca geoglyphs 31

Neanderthals 12, 188
 first finds 203
 fossils 165
 genome 347
 NEAR Shoemaker spacecraft 338
 nebulae
 Halley's explanation 130
 planetary 54
 nebular hypothesis 138, 149
 scientific 163
 nebular theory for solar system
 144, 145
 Neddermeyer, Seth 269
 negative numbers 42
 Neogene period 367
 neon 233, 359
 Ramsey 373
 neon lighting 229, 252
 Neoplatonism 38
 neoprene 263
 Neptune 364
 discovery of 198
 orbit 100, 101
 Voyager 2 space probe 323
 Neptunism 153
 neptunium 272, 359
 Nereid 280
 Neri, Antonio 97
 nerve cell
 structure 209
 see also neuron
 nerve fibers 161
 nerve impulse 287
 nerve messages, Galvani 370
 nervous system, Golgi 370
 Neumann, Franz Ernst 189
 neurology, Duchenne 369
 neuron 214
 coining of term 227
 see also nerve cell
 neuroscience 229
 neutrino oscillation 338
 neutrinos 296, 340, 342
 proposed 263
 neutron star 54, 55, 268, 296
 neutrons 250, 251, 266, 267
 and cosmology 344
 discovery 264
 in nuclear reactions 268
 Standard Model 304
 new algebra 92
 New Horizons 342
New Methods of Celestial Mechanics
 227
 New York Botanical Gardens 227
 Newcomen, Thomas 127, 129
 biography 372
 Newcomen engine 170
 Newlands, John 209
 Newsham, Richard 134
 newton, definition 352
 Newton, Isaac **118**, 120, 121, 372
 and light 116
 and orbits 100
 and planetary motion 117
 and sound 124, 125
 increasing fame 126
 Optiks 127
 prism experiment 112
 rainbow 113
 size of Earth 138
 star catalog 129
 telescope 112, 113
 Newton's laws of motion **120–21**,
 355
 NiCad batteries 323
 niche, ecological 253
 Nicholas of Cusa 68, 69, 70
 Nicholas of Oresme 65, 67
 Nicholson, William 167
 Nicolle, Charles 247
 Nicomachus 40

Nicot, Jean 82, 83
Nicotiana 82
 nicotine addiction 322
 Niépce, Joseph Nicéphore 186,
 187
 Niépce Brothers engine 173
 Nightingale, Florence 372
 Nile River 14
 source of 203
 Nilometer 18
 NiMH batteries 323
Niña 71
 niobium 168, 359
 nitrogen 359
 atom 250
 first isolated 152
 fixation 368
 nitrous oxide 152
 and global warming 326
 NMR [nuclear magnetic
 resonance] **278**
 Nobel, Alfred 210, 372
 noble gases 229, 232, 233, 358
 Noether, Emmy 372
 non-Euclidean geometry 27, 183,
 201
 Nooth, John Mervyn 151
 Norman, Robert 80
 North and South Poles 80, **94**
 Northern Lights 288
 Northwest Passage 181, 198
 Novoselov, Kostya 340
 Noyce, Robert 288
 NSFnet 322
 nuclear accident
 Chernobyl 320
 Three Mile Island 310
 nuclear bomb 268
 nuclear fission 270, **271**
 nuclear fusion **323**
 in stars 54, 55
 nuclear magnetic resonance
 (NMR) **278**
 nuclear power 235
 nuclear reactor **310**
 Fermi 370
 first 273
 nuclear test, postwar 278
 nucleic acid 284
 nuclein 214
 nucleolus 194, 195
 nucleon 266
 nucleotides (DNA) 254
 nucleus (atom) 250, 251
 and radioactivity 266, 267
 first signs 246
 named 248
 nucleus (cell) 194, 195
 and heredity 223
 naming 189
 nuclide 266
 number theory 39, 93
 early 29
 numbers
 development of **49**
 early systems 24
 systems of 46
 numerals 368
 nutation, Earth's 142
 nylon 261, 268, 270

O

O'Glacan, Neil 103
 oars for boats 19
 Oberon 157
 obesity hormone **329**
 occupational medicine 126
 ocean floor 281
 oceanography, story of **292–93**

Ockham, William of 373
 Ockham's razor 65
 octant 137
 occultation, planetary 139
 Odierna, Giovanni 105
 Odling, William 202
 Ørsted, Hans 182, 186
 Ogawa, Seiji 324
 ohm, definition 352
 Ohm, Georg Simon 373
 Ohm's Law 356, 373
 Ohno, Susumu 303
 Okabe, Masaru 332
 okapi 237
 Olbers, Heinrich 183
 biography 373
 Olbers, Wilhelm 169
 Olbers' paradox 183, 373
 Oldham, Richard 242
 Oldowan tools 12
 oligonucleotide synthesis 347
 Oliver, George 229
 omnibus 186
On the Origin of Species 206, 207
 oncogenes 314
 Onnes, Heike Kamerlingh 246, 247
 biography 371
 Oort, Jan 265, 280, 281
 Oort Cloud 281
 Opaque era, Universe 345
 Oparin, Alexander 256
 open-heart surgery 282, 368
 ophthalmoscope 91
 Oppenheimer, Robert 373
 optical fibers 301
 optics 50, 51, 60, 61
 Alhazen 368
 Brewster 369
Optiks 127
Opus Majus 60
 oral tradition 316
 Orbel, Geoffrey 279
 orbital electrons 250, 251
 orbital periods 101
 orbital speeds 100
 orbits, planetary 96, **100–101**, 364
 orders, taxonomy 360
 Ordnance Survey 141
 Ordovician period 367
 organ transplantation 368
 organelles 194, 300
 organic chemistry 187
 Oribasius of Pergamum 38, 39
 origin of life, Miller-Urey
 experiments 282, 283
 Orion Nebula 96, 97, 151
 ornithology 118
 ornithopter 240
 orreries 127
 Ørsted, Hans Christian 181
 biography 373
 Ortelius, Abraham 86, 93
 Osborne, Henry 236
 Osborne 1 portable computer 312,
 313
 oscillation, laws of 104
 oscilloscope 232
 Østerberg, Gustav 269
 Ostrom, John 297
 Ōsumi 300
 Oswalt, Terry 333
 Otis, Elisha 203
 Otto, Nikolaus 171, 216
 biography 373
 Otto cycle 373
 Oughtred, William 99, 103, 184
 biography 373
 oviduct 83
 Owen, Richard 187, 196, 197
 biography 373
 oxygen 359
 discovery **152**

Priestley 373
 respiration 195
 ozone layer 304
 and global warming 326–27
 depletion 319
 hole growth 328
 measuring 287
 Montreal Protocol 321
 naming 193

P

p-orbital 251
 P-waves 242, 243
 Pacioli, Luca 71
 paddle steamer 157
 Page, Larry 332
 Paine, George 263
 Paleogene period 367
 paleontology 164
 Cope 369
 Paleozoic era 366
 Palermo, Gianpiero 328
 palladium 359
 discovery 172
 Pallas, Peter Simon 152, 169
 Pallas's cat 152
 Palmer, Nathaniel 181
 Palmieri, Luigi 222
 palynology 253
 Pangaea 252
 pangenesis 237, 374
 pangenesis 143, 211
 Panhard car 227
 Pantridge, Frank 296
 paper, invention of 31, 37
 paper money 46
 Papin, Denis 117, 122, 170
 biography 373
 Pappus of Alexandria 38
 papyri 22
 Paracelsus 72, 86
 biography 373
 parachute 71, 156, 157
 paraffin 198, 199, 200
 parallax 95
 astronomy 193
 Paré, Ambroise 83
 parheliometer 146
 Parkes, Alexander 203, 260, 261
 parkesine 203, 260, 261
 Parkinson, James 161
 Parkinson, John 103
 botanical work 104
 Parkinson's disease 161
 Parry, William 181
 partial difference equations 142
 partial pressures, law of 168
 particle accelerator 244, 264, 278,
 279
 particle-wave duality 235, 257
 Pascal, Blaise 105, 106, **107**, 184,
 373
 early works 104
 pascal, definition 352
 Pascal-Fermat theory 107
 Pascal's Law 107
 Pascal's theorem 104
 Pascal's Triangle 52, 73, 107
 Pascaline 105, 184
 Pasteur, Louis **214**, 373
 anthrax 221
 fermentation 203
 puerperal fever 209
 rabies 223
 pasteurization 214
 patents 68
 pathology 374
 Patterson, Claire 286
 Paul of Aegina 42